

III.

$$(a) \quad X^T X \beta = X^T y \Rightarrow X^T X (X^T X)^{-1} X^T y = X^T y$$

$$(X^T X)^{-1} X^T y =: x_p ; \text{ particular solution}$$

$$\text{Note that } \hat{\beta}_h = x_p + x_h$$

where x_h is homogeneous solution s.t. $X^T X x_h = 0$.

$$X^T X (I - (X^T X)^{-1} X^T X) = 0$$

$$\text{So, } (I - (X^T X)^{-1} X^T X) h = x_h$$

$$\hat{\beta}_h = x_p + x_h$$

$$= (X^T X)^{-1} X^T y + (I - (X^T X)^{-1} X^T X) h, \quad h \in \mathbb{R}^p$$

□

$$(b) \text{ Let } \tilde{\beta} = A\gamma \text{ s.t. LUE.}$$

$$\mathbb{E}[\tilde{\beta}] = A X \beta = \beta \Rightarrow AX = I.$$

$$\text{rank}(I) = \text{rank}(AX) \leq \text{rank}(X) = r \neq p$$

$\therefore \nexists$ LUE

□

$$e = y - X\hat{\beta}$$

$$= y - X(X^T X)^{-1} X^T y - \underbrace{X(I - (X^T X)^{-1} X^T X)}_{=0} h$$

$$= (I - X(X^T X)^{-1} X^T) y$$

$$e^T e = y^T (I - X(X^T X)^{-1} X^T) y \quad \because I - X(X^T X)^{-1} X^T : \text{idempotent}$$

$$\text{tr}(I - X(X^T X)^{-1} X^T) = n - r \quad \because \text{tr}(X(X^T X)^{-1} X^T) = \text{tr}(X) = r$$

$$\frac{1}{2} \mu^T A \mu = 0$$

$$\frac{e^T e}{\sigma^2} \sim \chi^2(n-r)$$

$$\frac{e^T e}{\sigma^2} = \chi^2_{n-r} \Rightarrow \frac{e^T e}{\chi^2_{n-r}} = \sigma^2$$

$$\sigma^2 \in \left[\frac{e^T e}{\chi^2_{n-r, 1-\alpha/2}}, \frac{e^T e}{\chi^2_{n-r, \alpha/2}} \right]$$

□

2.

$$(a) \hat{SE}(\hat{\beta}) = s \sqrt{v_{jj}}, \quad v_{jj} = \text{diagonal of } (X^T X)^{-1}$$

$$T_j = \frac{\frac{\hat{\beta}_j - \beta_{0j}}{s \sqrt{v_{jj}}}}{\sqrt{\frac{s^2}{\sigma^2}}} \sim N(0,1) \quad \perp \quad \sim \pm n-p$$

$$\sim \chi^2(n-p)$$

$$(X^T X)^{-1} X^T (I - H_X) = 0$$

□

$$(b) P(|T_j| > t_{\alpha/2p, n-p}) = \frac{\alpha}{p}$$

$$P\left(\bigcup_{j=1}^p |T_j| > t_{\alpha/2p, n-p}\right)$$

$$\leq \sum_{j=1}^p P(|T_j| > t_{\alpha/2p, n-p})$$

$$= \frac{\alpha}{p} \cdot p = \alpha$$

□

$$(c) S = B B^T \Rightarrow S^{-1} = (B^T)^{-1} B^{-1}$$

$$v = B^{-1}(\hat{\beta} - \beta_0)$$

$$u = B^T e_j$$

$$T_j^2 = \frac{(e_j^T (\hat{\beta} - \beta_0))^2}{e_j^T S e_j} \leq \frac{(\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0)}{\uparrow \text{Cauchy-Schwarz}}$$

The equality holds when $u \propto v$

$$\Rightarrow B^T e_j \propto B^T (\hat{\beta} - \beta_0)$$

$$\Rightarrow e_j \propto S^{-1} (\hat{\beta} - \beta_0)$$

□

(d)

$$T_j^2 \leq (\hat{\beta} - \beta_0)^T \underbrace{S^{-1}}_{S^2(X^T X)} (\hat{\beta} - \beta_0)$$

$$= \frac{(\hat{\beta} - \beta_0)^T X^T X (\hat{\beta} - \beta_0)}{S^2}$$

$$\Rightarrow \frac{(y - X\beta_0)^T H_X (y - X\beta_0) / p}{(y - X\beta_0)^T (I - H_X) (y - X\beta_0)} \sim F_{p, n-p}$$

$$H_X (I - H_X) = 0$$

$$\Rightarrow (\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p}$$

$$P((\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p}) = 1 - \alpha$$

$$P\left(\bigcap_{j=1}^p \{T_j^2 \leq p F_{p, n-p}\}\right)$$

$$\geq P((\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p})$$

$$= 1 - \alpha$$

□

[3]

$$\sum_{i=1}^T y_i = \mathbf{1}^T \mathbf{y}$$

$$y_T = \mathbf{1}^T \mathbf{y} + \mathbf{x}_T^T \boldsymbol{\beta} + \varepsilon_T$$

$$\hat{\boldsymbol{\beta}}^{BLUE} = (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1} \mathbf{y}$$

$$\begin{aligned} \text{cov}(y_T, \mathbf{y}) &= \text{cov}\left(\sum_{i=1}^T y_i + \mathbf{x}_T^T \boldsymbol{\beta} + \varepsilon_T, \mathbf{y}\right) \\ &= \text{cov}(\mathbf{1}^T \mathbf{y} + \mathbf{1}^T \boldsymbol{\varepsilon}, \mathbf{y}) \\ &= \mathbf{1}^T \mathbf{V} \end{aligned}$$

$$\text{Var}(\mathbf{y}) = \mathbf{V}$$

$$\begin{aligned} \mathbb{E}(y_T) &= \mathbb{E}(\mathbf{1}^T \mathbf{y} + \mathbf{x}_T^T \boldsymbol{\beta} + \varepsilon_T) = \mathbf{1}^T \mathbf{X} \boldsymbol{\beta} + \mathbf{x}_T^T \boldsymbol{\beta} \\ &= (\mathbf{1}^T \mathbf{X} + \mathbf{x}_T^T) \boldsymbol{\beta} \end{aligned}$$

$$\begin{aligned} \hat{y}_T^{BLUP} &= (\mathbf{1}^T \mathbf{X} + \mathbf{x}_T^T) \hat{\boldsymbol{\beta}}^{BLUE} + \mathbf{1}^T \mathbf{V} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}^{BLUE}) \\ &= \mathbf{x}_T^T \hat{\boldsymbol{\beta}}^{BLUE} + \mathbf{1}^T \mathbf{y} \\ &= (\mathbf{1}^T + \mathbf{x}_T^T (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1}) \mathbf{y} \end{aligned}$$

$$\text{Let } \hat{\mathbf{y}}_T = \mathbf{a}^T \mathbf{y}$$

$$\Rightarrow \mathbb{E} \hat{\mathbf{y}}_T = \mathbb{E} y_T$$

$$\Rightarrow \mathbf{a}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{1}^T \mathbf{X} \boldsymbol{\beta} + \mathbf{x}_T^T \boldsymbol{\beta}$$

$$\Rightarrow \mathbf{a}^T \mathbf{X} = \mathbf{1}^T \mathbf{X} + \mathbf{x}_T^T \text{ \& } \mathbb{E} \hat{\mathbf{y}}_T = \mathbf{1}^T \mathbf{X} \boldsymbol{\beta} + \mathbf{x}_T^T \boldsymbol{\beta} : \text{Unbiased.}$$

$$\text{Var}(a^T y - y_T)$$

$$= \text{Var}((a - 1)^T y + x_T^T \beta + e_T)$$

$$= (a - 1)^T V (a - 1) + 0 + \sigma_T$$

$$\text{Var}(\hat{y}_T - y_T)$$

$$= \text{Var}((1^T + x_T^T (X^T V^{-1} X)^{-1} X^T V^{-1}) y - 1^T y - x_T^T \beta - e_T)$$

$$= x_T^T (X^T V^{-1} X)^{-1} x_T$$

$$\text{SMSE}(\hat{y}_T) - \text{SMSE}(y_T)$$

$$= (a - 1)^T V (a - 1) - x_T^T (X^T V^{-1} X)^{-1} x_T$$

$$= (a - 1)^T V (a - 1) - (a - 1)^T X (X^T V^{-1} X)^{-1} X^T (a - 1)$$

$$= (a - 1)^T (V - X (X^T V^{-1} X)^{-1} X) (a - 1) \geq 0$$

$$V^{1/2} [I - A(A^T A)^{-1} A^T] V^{1/2},$$

$$A = V^{1/2} X \quad \hookrightarrow \text{p.d.}$$

So, $\hat{y}_T^{\text{BLUP}}$ is BLUP.



5.

$$(a) \log f(y_i | x_i) = y_i \log \mu_i + (1 - y_i) \log (1 - \mu_i) \\ = y_i \log \frac{\mu_i}{1 - \mu_i} + \log (1 - \mu_i)$$

$$l(\beta) = \sum (y_i \theta_i - b(\theta_i))$$

$$\Rightarrow \theta_i = \log \frac{\mu_i}{1 - \mu_i}$$

$$-b(\theta_i) = \log (1 - \mu_i) = \log \left(\frac{1}{1 + e^{\theta_i}} \right)$$

$$\Rightarrow b(\theta_i) = \log (1 + e^{\theta_i})$$

$$(b) \frac{\partial l_i}{\partial \mu_i} = \frac{y_i}{\mu_i} - \frac{1 - y_i}{1 - \mu_i} = \frac{y_i - \mu_i}{\mu_i (1 - \mu_i)}$$

$$\mu_i = \Phi(x_i \beta) \Rightarrow \frac{\partial \mu_i}{\partial \beta} = \phi(x_i \beta) x_i$$

$$S(\beta) = \sum_{i=1}^n \frac{\partial l}{\partial \mu_i} \frac{\partial \mu_i}{\partial \beta} = \sum_{i=1}^n \frac{y_i - \Phi(x_i \beta)}{\Phi(x_i \beta) (1 - \Phi(x_i \beta))} \phi(x_i \beta) x_i$$

$$(c) I(\beta) = \text{Var}(S(\beta)) \\ = \sum_{i=1}^n \left(\frac{\phi(x_i \beta) x_i}{\Phi(x_i \beta) (1 - \Phi(x_i \beta))} \right)^2 \overset{\Phi(x_i \beta) (1 - \Phi(x_i \beta))}{\parallel} \text{Var}(y_i) \\ = \sum_{i=1}^n \frac{\phi(x_i \beta)^2 x_i^2}{\Phi(x_i \beta) (1 - \Phi(x_i \beta))}$$

(d) Under H_0 , $\mathbb{E}(0) = 0.5$, $\phi(0) = \frac{1}{\sqrt{2\pi}}$

$$S(0) = \sum_{i=1}^n \frac{y_i - 0.5}{0.25} \frac{1}{\sqrt{2\pi}}$$

$$= \frac{4}{\sqrt{2\pi}} \sum_{i=1}^n (y_i - 0.5)$$

$$= \frac{8\sqrt{n}}{\sqrt{2\pi}}$$

$$I(0) = \sum_{i=1}^n \frac{\left(\frac{1}{\sqrt{2\pi}}\right)^2}{0.25} x_i^2 = \frac{4}{2\pi} \sum_{i=1}^n x_i^2 = \frac{2n}{\pi}$$

$$Z = \frac{S(0)}{\sqrt{I(0)}} = \frac{8\sqrt{n} / \sqrt{2\pi}}{\sqrt{2n/\pi}} = \frac{8}{2} = 4 > 1.96$$

$\therefore$ reject H_0 $\blacksquare$